Optimized Deep Learning and Sentiment-Aware Decision Framework for Personalized Stock Market Investment Support

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Abstract. Stock price prediction is challenging due to the complex and dynamic nature of financial markets influenced by both historical trends and external factors like news sentiment and investor behavior.
Our project presents a consolidated stock market decision framework that integrates three components: price forecasting, sentiment evaluation, and personalized recommendation. A Long Short-Term Memory (LSTM) network is utilized for time series prediction, and its hyperparameters are optimized using a Genetic Algorithm (GA) to enhance convergence and reduce prediction error. Analysis of financial news sentiment is performed using a domain-adapted transformer model (FinBERT). Furthermore, a profile-aware rule-based recommendation module translates predictive insights into actionable investment suggestions.
Experimental validation is performed on Amazon (AMZN) stock data. Comparative evaluation between baseline LSTM and GA-optimized LSTM illustrates improved error metrics and greater forecast consistency. Sentiment analysis provides contextual reinforcement to price trends, while the recommendation module prioritizes interpretability and decision stability. The findings indicate that combining optimized deep learning models with sentiment awareness and transparent decision logic can offer a practical framework for retail investment support.

Keywords: Stock Price Prediction · LSTM · Genetic Algorithm · Sentiment Analysis · FinBERT · Rule-Based Recommendation System

1 Introduction

Financial markets operate as complex and highly volatile systems where asset prices evolve under the combined influence of structured numerical data and unstructured information streams. Accurate prediction in such environments is inherently difficult due to rapid variations and sensitivity to macroeconomic, geopolitical, and behavioral factors [1, 2].

Traditional investment strategies relied on manual interpretation of charts and financial indicators. However, the growing availability of large-scale data and computational resources has encouraged the adaptation of machine learning

and deep learning techniques for automated forecasting [3]. Among these, Long Short-Term Memory (LSTM) networks have gained prominence for modeling sequential financial data, as they are capable of retaining relevant historical information over extended time horizons [4].

Despite their strengths, LSTM models are sensitive to architectural and training parameters such as sequence length, hidden units, and learning rate. Suboptimal parameter selection can significantly degrade forecasting performance [5]. Evolutionary search strategies, including Genetic Algorithms (GA), provide a systematic mechanism for identifying improved hyperparameter configurations [6].

Beyond numerical price patterns, market behavior is strongly shaped by investor sentiment and financial news. Domain-specific language models such as FinBERT enable structured interpretation of financial text by capturing contextual nuances specific to economic reporting [7]. Integrating such sentiment signals with quantitative models can enhance market understanding [8].

Ultimately, accurate predictions alone do not provide a complete solution for retail investors, who possess varying levels of risk tolerance and capital. There is a critical need for personalized guidance that translates predictive insights into actionable advice. This paper proposes an integrated methodology that combines an LSTM-GA hybrid forecasting model, FinBERT-based sentiment analysis, and a profile-driven recommendation system to provide comprehensive decision support for the modern investor.

2 Literature Survey

Financial markets strongly affect business, jobs, and technology. Investors primarily use fundamental analysis and technical analysis approaches before making a decision [9]. With the advancement in technology in recent years, stock price prediction is no longer done manually. Several machine learning algorithms are implemented now to forecast stock prices and trends. At the beginning, regression models were used, but later Artificial Neural Networks and Support Vector Machines became popular due to their capability to learn complex patterns in stock data [10].

Day-wise past stock prices were collected from the Bombay Stock Exchange. The model used Date and Close Price for training. LSTM was the superior choice for stock market analysis because of its capability to handle long-term dependencies, solving vanishing gradient problems, contextual persistence, and superiority over regression [11]. LSTM has several technical advantages such as differentiable analog gates (implemented with sigmoid functions) and the capacity to read and remember intermediary context for varying periods without using activation functions for each recurring component. This makes LSTM a strong choice for the volatile nature of the stock market [12].

Optimization is important because hyperparameters significantly influence the prediction performance and training parameters of deep learning models. This study uses Bayesian Optimization along with the LSTM network [13]. Ge-

netic Algorithm (GA) provides a simple yet robust evolutionary approach to hyperparameter tuning. GA are specifically effective for complex search spaces due to their ability to manage a vast domain of variables simultaneously. GA are particularly well suited for LSTM networks because of the intricate dependencies between the LSTM architecture and its performance [14].

Researchers have found that financial news predicts stock market volatility better than closing prices alone [15]. Sentiments have an asymmetric impact on market prices. Negative sentiments have the strongest influence on market movement, while an increase in positive sentiment correlates with ascending prices and a decrease in market volatility [16]. FinBERT is a large language model specifically customized for the financial domain. It is designed to overcome the limitations of generic NLP models and is used for sentiment analysis of financial documents. Investment recommendation systems work as decision support tools that perform analysis of user behaviour and data to provide personalized advice based on specific needs. IF–THEN rules give a clear reasoning mechanism, allowing users to see which inputs led to a specific recommendation. The proposed methodology in the research is a hybrid approach that uses K-means clustering to organize investment types and neuro-fuzzy techniques to map those types to an individual's psychological and environmental priorities[17].

3 Methodology

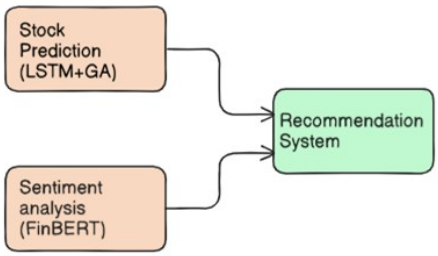

Fig. 1. Overall working of the system

The approach adopted within our proposed system consists of three components: (1) stock price prediction, (2) financial sentiment analysis, and (3) profile-driven recommendation system. These components work together to deliver a sentiment-aware market insight along with user-specific investment recommendations. The overall working of the system is demonstrated in Fig. 1.

3.1 Stock Price Forecast

Yahoo Finance (yfinance), a Python-based financial data library, is used to obtain historical stock market data. The collected dataset includes Open, High,

Low, Close, and Volume (OHLCV) attributes, among which the closing price is selected as the prediction target. Missing values in the time-series data are handled using forward and backward filling techniques to maintain continuity.

To address variations in stock price ranges and improve learning stability, Min–Max normalization is applied using

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}} \tag{1}$$

For stock price prediction, a Long Short-Term Memory (LSTM) network is employed and implemented from scratch using the PyTorch framework. All four LSTM components, namely the input gate, forget gate, output gate, and cell state, are explicitly constructed.

The forget gate is defined as:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \tag{2}$$

while the input gate is given by:

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \tag{3}$$

The candidate cell state is computed as:

$$\tilde{C}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \tag{4}$$

and the cell state update follows:

$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t \tag{5}$$

The output gate is expressed as:

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \tag{6}$$

producing the hidden state:

$$h_t = o_t \cdot \tanh(C_t) \tag{7}$$

Model parameters are initialized using Xavier initialization, defined as:

$$W \sim U\left(-\sqrt{\frac{6}{n_{in} + n_{out}}}, \sqrt{\frac{6}{n_{in} + n_{out}}}\right) \tag{8}$$

and are progressively updated during training. Several hyperparameters, including the number of epochs, number of neurons, sequence length, dropout rate, and learning rate, are specified prior to training. Initial default values are chosen to enable faster convergence, after which a Genetic Algorithm (GA) is applied to optimize these parameters by minimizing the prediction loss. The overall methodology adopted in this work is illustrated in Fig. 2.

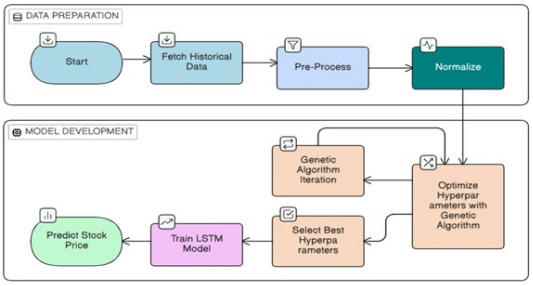

Fig. 2. Stock price prediction workflow

3.2 Sentiment Analysis

Financial sentiment is incorporated to capture qualitative factors influencing market behavior. A labeled dataset of financial news was constructed using the Loughran–McDonald sentiment dictionary, which identifies positive, negative, and neutral terms specific to financial text. News sentences were collected from multiple sources and preprocessed by converting text to lowercase and removing punctuation, numbers, and stopwords.

The final dataset consists of approximately 1,200 sentences, categorized into Positive (420), Negative (390), and Neutral (390) classes. To improve labeling reliability, dictionary-based annotations were supplemented with manual verification of ambiguous samples. The dataset was split into training and testing sets in a 90–10 ratio, and the training data was used to fine-tune the FinBERT model.

FinBERT, a domain-adapted transformer model, leverages a self-attention mechanism defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{9}$$

where Q, K, and V denote query, key, and value matrices, and d_k represents the key dimensionality. The softmax function converts attention scores into probabilities:

$$\text{softmax}(x_i) = \frac{e^{x_i}}{\sum_j e^{x_j}} \tag{10}$$

For inference, real-time financial news is retrieved from Finnhub and classified into sentiment categories. Sentiment scores are mapped using predefined thresholds: values greater than 0.1 indicate positive, values less than -0.1 indicate negative, and intermediate values are classified as neutral.

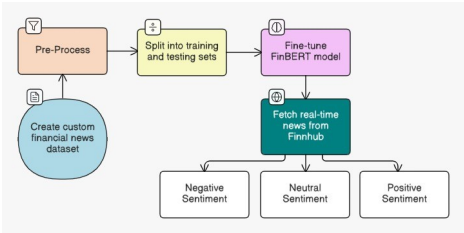

Fig. 3. Sentiment Analysis workflow

As illustrated in Fig. 3, the workflow consists of dataset construction, preprocessing, model fine-tuning, and real-time sentiment classification. This pipeline enables the integration of contextual sentiment signals with quantitative market data for downstream decision-making.

3.3 Recommendation System

The recommendation module integrates technical indicators, sentiment signals, and user-specific constraints to generate personalized investment decisions. User inputs include age, market type, and investment amount, which are used to model risk tolerance.

Market trends are captured using standard technical indicators such as Simple Moving Average (SMA) and Exponential Moving Average (EMA), defined as:

$$SMA = \frac{1}{n}\sum_{i=1}^{n} P_i \tag{11}$$

$$EMA_t = \alpha P_t + (1-\alpha)EMA_{t-1}, \quad \text{where} \quad \alpha = \frac{2}{n+1} \tag{12}$$

Market momentum is measured using the Relative Strength Index (RSI):

$$RSI = 100 - \frac{100}{1+RS} \tag{13}$$

where RS denotes the ratio of average gains to losses.

User risk compatibility (R) is determined using age-based constraints and stock volatility classification (low, medium, high risk). Stocks violating user risk constraints are penalized or excluded.

Final recommendations are generated through a three-step process:

- **Filtering:** Stocks are filtered based on risk compatibility and predicted price direction.
- **Scoring:** Each stock is assigned a composite score using the defined function.
- **Ranking:** Stocks are ranked in descending order of normalized scores, and top candidates are labeled as Buy, Hold, or Sell based on predefined thresholds.

Algorithm 1 Recommendation Generation Procedure

Input: Stock data D, sentiment score S, user profile U
Output: Ranked stock recommendations

1. Compute SMA, EMA, and RSI from D
2. Determine technical signal T
3. Classify stock risk using volatility
4. Apply user risk constraints
5. Compute Score $= w_1 T + w_2 S + w_3 R$
6. Filter incompatible stocks
7. Rank stocks based on score
8. Assign Buy/Hold/Sell labels
9. Return top recommendations

Algorithm 1 outlines a rule-based recommendation procedure that integrates technical indicators, sentiment scores, and user profile constraints. It computes market signals, classifies risk, and derives a weighted confidence score to filter, rank, and assign actionable labels (buy/hold/sell) to stocks, ultimately returning the top recommendations.

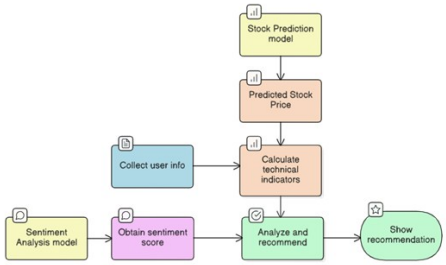

Fig. 4. Recommendation system workflow

As illustrated in Fig. 4, the proposed framework operates as a multi-source decision pipeline integrating predictive modeling, sentiment analysis, and rule-based recommendation. Let D_t denote historical stock data up to time t, and U represent user-specific attributes.

A stock prediction model $\mathcal{M}$ (LSTM–GA) is first applied to D_t to estimate the future price.

In parallel, a sentiment analysis model (FinBERT) processes financial text T to generate a sentiment score:

$$S = \mathcal{S}(T) \tag{14}$$

Subsequently, technical indicators $\mathbf{I} = f(D_t)$ are computed from historical price data. The recommendation module then performs a joint analysis over historical data, $\mathbf{I}$, S, and U. Specifically, price direction is inferred from $\mathbf{I}$, risk levels are derived from user constraints (U) and stock volatility, and a confidence score is computed as:

$$C = g(\mathbf{I}, S) \tag{15}$$

A filtering function $\mathcal{F}$ selects stocks satisfying both predicted direction and user-specific risk constraints:

$$D' = \mathcal{F}(D_t, U, \mathbf{I}, S) \tag{16}$$

4 Results

Amazon (AMZN) stock was selected for evaluation due to its high market capitalization, substantial trading volume, and extensive coverage in financial news media. These characteristics make it a suitable candidate for evaluating both price prediction models and sentiment analysis techniques.

The stock price prediction framework was evaluated using a baseline Long Short-Term Memory (LSTM) model and a hybrid LSTM model optimized using a Genetic Algorithm (GA). Model performance was assessed using three commonly used regression metrics: Mean Absolute Percentage Error (MAPE), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE).

Using historical stock data obtained from Yahoo Finance, the baseline LSTM model predicted a future price of \$220.14, compared to the current price of \$245.80, indicating a projected decline of \$25.60 (-10.42%). After applying Genetic Algorithm–based hyperparameter optimization, the predicted price improved to \$224.12, reducing the projected decline to \$21.68 (-8.82%).

The optimized hyperparameters identified by the GA include a sequence length of 20, 78 hidden units, and two LSTM layers, which together improved the model's predictive performance.

4.1 Stock Price Prediction

The predictive performance of the baseline LSTM model and the GA-optimized LSTM model was compared using standard regression metrics.

Table 1. Comparison between LSTM and GA-Optimized LSTM

Method	MAPE	RMSE	MAE
LSTM (without GA)	2.25%	7.18	5.70
LSTM (with GA)	1.92%	5.47	4.33

The results demonstrate that Genetic Algorithm–based optimization improves the predictive performance of the LSTM model. The GA-optimized LSTM achieves lower MAPE, RMSE, and MAE values, indicating better model convergence and an enhanced ability to capture nonlinear patterns present in stock price movements.

4.2 Sentiment Analysis

i) Sentiment Classification Results for Amazon (FinBERT)

FinBERT was used to classify financial news articles into three sentiment categories: Positive, Neutral, and Negative. As a domain-specific model trained on financial text, it effectively captures contextual patterns in financial news.

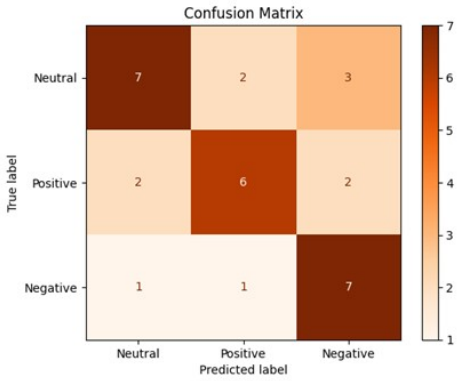

Fig. 5. Confusion Matrix for AMZN (Neutral / Positive / Negative)

The confusion matrix shows the relationship between true and predicted sentiment labels, indicating that FinBERT generates reliable sentiment signals for use in investment decision-support systems.

Table 2. Evaluation of Sentiment Classification

Class	Precision	Recall	F1-Score	Support
Neutral	0.70	0.58	0.64	12
Positive	0.67	0.60	0.63	10
Negative	0.58	0.78	0.67	9

ii) Mapping Sentiment to Trading Signals The classified sentiment outputs were further mapped to corresponding trading signals, including Buy, Hold, and

Sell. While sentiment signals alone cannot fully explain stock price movements, they provide useful directional insights when combined with technical indicators and predictive models.

Table 3. Evaluation of Trading Signals

Class	Precision	Recall	F1-Score	Support
Hold	0.73	0.80	0.76	10
Buy	0.67	0.60	0.63	10
Sell	0.70	0.70	0.70	10

The results indicate that sentiment-derived trading signals provide a meaningful complementary signal for investment decision support. In particular, the model demonstrates strong performance in identifying hold signals, which are important for preventing unnecessary trading during stable market conditions.

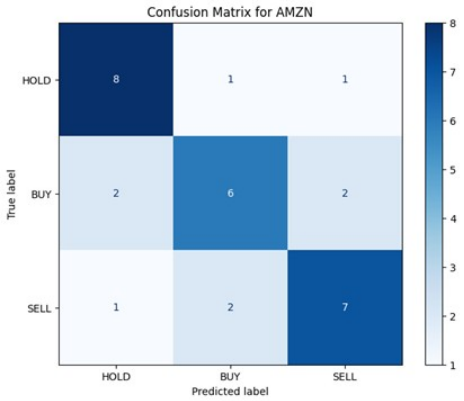

Fig. 6. Confusion Matrix for AMZN Trading Signals (Hold / Buy / Sell)

4.3 Recommendation System

The recommendation system was evaluated through backtesting on 190 Amazon (AMZN) trading instances collected over a one-year period. These samples were excluded from the model training phase to ensure an unbiased assessment of the recommendation module.

The evaluation results indicate that the proposed recommendation framework achieves stable performance across all recommendation categories, with strong performance in identifying hold signals that help avoid unnecessary trading during stable market conditions.

Table 4. Evaluation of Recommendation System

Class	Precision	Recall	F1-Score	Support
Buy	0.76	0.72	0.74	62
Hold	0.82	0.85	0.83	78
Sell	0.74	0.71	0.72	50
Macro Avg	0.77	0.76	0.76	190
Weighted Avg	0.78	0.78	0.78	190

The integration of GA-optimized LSTM price forecasts with FinBERT-based sentiment analysis improves recommendation accuracy by combining quantitative price patterns with qualitative market sentiment. The rule-based framework also ensures transparency and interpretability in the generated investment suggestions.

Overall, the backtesting results demonstrate that the proposed system effectively converts predictive insights into actionable investment recommendations while maintaining consistent performance across different market scenarios.

5 Conclusion

Forecasting stock market behavior remains challenging due to the interaction of quantitative price dynamics and qualitative factors such as financial news and investor sentiment. Relying solely on either source is often insufficient for reliable decision-making. This study proposes a hybrid decision-support framework integrating LSTM-based price prediction, Genetic Algorithm (GA)–based hyperparameter optimization, FinBERT-driven sentiment analysis, and a profile-based recommendation module.

Experimental results show that the GA-optimized LSTM model achieves improved prediction accuracy and stability compared to the baseline model. The inclusion of sentiment signals provides additional contextual insight, while the recommendation module translates predictive outputs into interpretable, user-specific investment guidance. However, the rule-based design limits adaptability to dynamic market conditions and does not explicitly support short-term trading strategies.

Acknowledgement: The authors acknowledge the use of ChatGPT (OpenAI) for language editing support. All generated content was reviewed and verified by the authors, who assume full responsibility for the final manuscript.

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